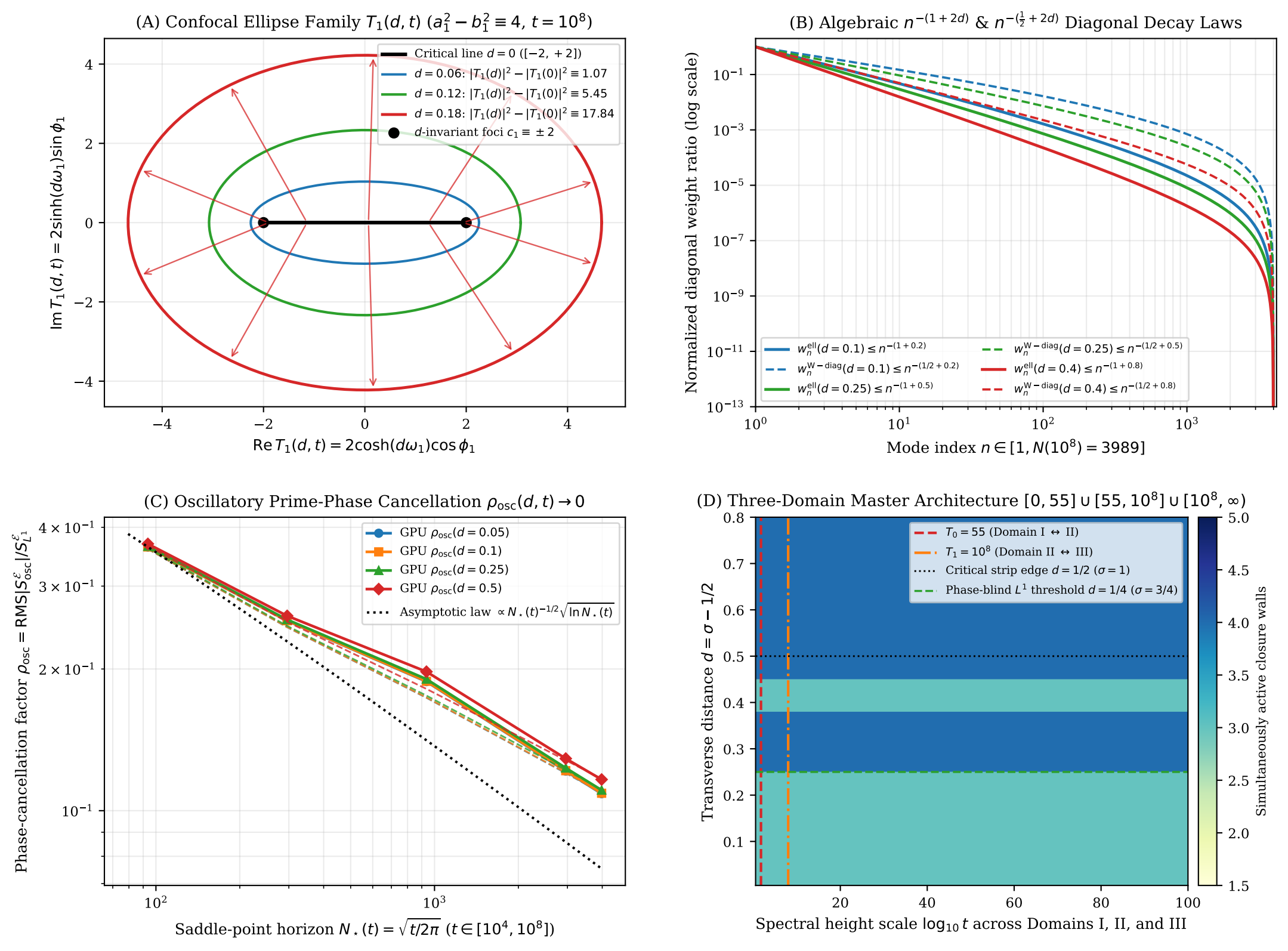

(A) Confocal Ellipse Family $T_1(d,t)$ ($a_1^2 - b_1^2 \equiv 4$, $t = 10^8$)
Critical line $d = 0$ ([−2, +2])
$d = 0.06$: $|T_1(d)|^2 - |T_1(0)|^2 \equiv 1.07$
$d = 0.12$: $|T_1(d)|^2 - |T_1(0)|^2 \equiv 5.45$
$d = 0.18$: $|T_1(d)|^2 - |T_1(0)|^2 \equiv 17.84$
d-invariant foci $c_1 \equiv \pm 2$
$\mathrm{Im}\, T_1(d,t) = 2\sinh(d\omega_1)\sin\phi_1$
$\mathrm{Re}\, T_1(d,t) = 2\cosh(d\omega_1)\cos\phi_1$
(B) Algebraic $n^{-(1+2d)}$ & $n^{-(\frac{1}{2}+2d)}$ Diagonal Decay Laws
Normalized diagonal weight ratio (log scale)
$w_n^{\mathrm{ell}}(d=0.1) \le n^{-(1+0.2)}$
$w_n^{\mathrm{W-diag}}(d=0.1) \le n^{-(1/2+0.2)}$
$w_n^{\mathrm{ell}}(d=0.25) \le n^{-(1+0.5)}$
$w_n^{\mathrm{W-diag}}(d=0.25) \le n^{-(1/2+0.5)}$
$w_n^{\mathrm{ell}}(d=0.4) \le n^{-(1+0.8)}$
$w_n^{\mathrm{W-diag}}(d=0.4) \le n^{-(1/2+0.8)}$
Mode index $n \in [1, N(10^8) = 3989]$
(C) Oscillatory Prime-Phase Cancellation $\rho_{\mathrm{osc}}(d,t) \to 0$
Phase-cancellation factor $\rho_{\mathrm{osc}} = \mathrm{RMS}|S_{\mathrm{osc}}^{\mathcal{E}}|/S_{L^1}^{\mathcal{E}}$
GPU $\rho_{\mathrm{osc}}(d=0.05)$
GPU $\rho_{\mathrm{osc}}(d=0.1)$
GPU $\rho_{\mathrm{osc}}(d=0.25)$
GPU $\rho_{\mathrm{osc}}(d=0.5)$
Asymptotic law $\propto N_\star(t)^{-1/2}\sqrt{\ln N_\star(t)}$
Saddle-point horizon $N_\star(t) = \sqrt{t/2\pi}$ ($t \in [10^4, 10^8]$)
(D) Three-Domain Master Architecture $[0,55] \cup [55, 10^8] \cup [10^8, \infty)$
$T_0 = 55$ (Domain I ↔ II)
$T_1 = 10^8$ (Domain II ↔ III)
Critical strip edge $d = 1/2$ ($\sigma = 1$)
Phase-blind L^1 threshold $d = 1/4$ ($\sigma = 3/4$)
Transverse distance $d = \sigma - 1/2$
Simultaneously active closure walls
Spectral height scale $\log_{10} t$ across Domains I, II, and III